

SAIF ULLAH ARSHAD

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EDUCATION

Bachelor of Computer Science, Information Technology University (ITU)
CGPA: 3.2 / 4 Relevant: Data Structures, Algorithms, Software Engineering, ML, OS

Aug 2023 – Jun 2027
Lahore, Pakistan

SKILLS

Languages: Python, JavaScript, TypeScript, C++
AI/ML: PyTorch, TensorFlow, OpenCV, CNNs, Scikit-learn
Tools: Git, Stripe API, RESTful APIs, JWT Auth, OOP

Frameworks: React.js, Node.js, Express.js, Tailwind CSS
Databases: PostgreSQL, MongoDB, MySQL

EXPERIENCE

Full Stack Engineer (Internship)
Nearpeer.org – EdTech Platform

Jun 2025 – Aug 2025
Lahore, Pakistan

- Identified performance bottlenecks in MongoDB architecture and proposed migration to PostgreSQL – redesigned schema across 12+ tables, cutting average query response time by **35%** and eliminating data integrity errors in enrollment records.
- Integrated Stripe payment gateway end-to-end (purchases, subscriptions, webhooks), enabling the platform's paid tier and unblocking **revenue collection for 5,000+ active learners**.
- Built JWT-secured auth and course enrollment APIs (Node.js/Express) handling **500+ daily requests** – collaborating with product and design teams to define role-based access control across student, instructor, and admin roles.
- Optimised React.js front-end with component-level lazy loading and Tailwind CSS, reducing page load time by **40%** on mobile – improving session retention on low-bandwidth connections.

Product Associate (Full-Time, Gap Year)
Out-Class – EdTech Platform

Nov 2022 – Jan 2024
Lahore, Pakistan

- Managed live platform serving **200+ learners across 50+ weekly sessions**; redesigned interactive content workflows achieving **30% uplift in engagement** and 25% faster turnaround – presenting performance reports to senior stakeholders to guide content strategy decisions.

PROJECTS

Urban Crash Analytics – ML Pipeline. [\[GitHub\]](#)

Python · XGBoost · GeoPandas · Scikit-learn · SMOTE

Built end-to-end ML pipeline predicting NYC road collision risk using spatial features across **500,000+ records**. Structured full workflow – data ingestion, cleaning, feature engineering, and class imbalance handling via SMOTE – translating raw geospatial data into actionable risk classifications with XGBoost.

EdTech Learning Platform with Stripe Integration. [\[View Project\]](#)

React · Node.js · Express · PostgreSQL · Stripe · Vercel

Built full-stack LMS with course enrollment, video streaming, and progress tracking serving **100+ test users**. Implemented Stripe subscriptions with webhook handling for real-time payment events. Migrated from MongoDB to PostgreSQL, cutting query load by **30%** and improving data reliability. Deployed on Vercel with CI via GitHub Actions.

Traffic Sign Classifier – CNN / PyTorch. [\[Live Demo\]](#) [\[GitHub\]](#)

PyTorch · OpenCV · Python · CNN

Engineered CNN achieving **92% accuracy** on **49,000+ image** GTSRB dataset – outperforming baseline models by 11%. Built complete pipeline: data augmentation, class-balanced sampling, and hyperparameter tuning. Deployed real-time web classifier with OpenCV for live image upload and prediction.

Real-Time Analytics Dashboard. [\[GitHub\]](#)

React · Chart.js · Node.js · PostgreSQL

Built interactive analytics dashboard with dynamic filters, date ranges, and CSV export for **50K+ row datasets**. Engineered PostgreSQL aggregation queries delivering results in **sub-200ms** – optimised with indexed views and connection pooling. Supports multi-metric comparison across configurable time windows.

EXTRA-CURRICULAR & ACHIEVEMENTS

- GDG × ITU Code Rush, 2024** – Competed in algorithmic programming contest; solved graph and DP problems under timed conditions against 100+ participants.
- Ethical Hacking Certificate – Udemy, 2024** – Completed 30+ hour course covering penetration testing, network security, and vulnerability assessment; applied secure coding practices in internship APIs.
- Kaggle – ML/AI Hackathon & Competitions, 2025** – Participated in ML/AI hackathon; earned Community Competitor, Code Uploader, and Python Coder badges through public notebook submissions and dataset challenges.